TITLE

Smart Glasses: An Intelligent Vision-Based Assistive Device for the Visually Impaired

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ABSTRACT

This paper presents the design and development of **VisionAssist**, a wearable assistive device in the form of smart glasses designed to enhance the spatial awareness of visually impaired individuals. Unlike traditional mobility aids, this system utilizes a dual-processor architecture: an **ESP32** for high-speed data handling and Wi-Fi/Bluetooth connectivity, and an **Arduino** (MCU) for peripheral management. The system integrates the **HuskyLens AI vision sensor** to perform real-time object recognition and face detection. Visual data is converted into clear, audible descriptions via a **PAM8403 audio amplifier** and a localized speaker system. Experimental results indicate that the device effectively identifies obstacles and environmental landmarks with high accuracy, providing a seamless audio-visual bridge that promotes independent mobility and safety.

1.INTRODUCTION

Visual impairment and age-related vision loss can greatly impact an individual's functionality in terms of safely moving about their environment. The visually impaired and elderly may experience problems such as avoiding obstacles, detecting objects, and properly moving about in a crowd or unknown environment, especially in reduced light conditions or during an emergency situation [1]. Conventional assistive devices, such as the white cane, lack ground-level awareness and cannot detect obstacles in front or above ground level, thus increasing the chances of accidents [2].

According to the World Health Organization, millions of people worldwide are living with visual impairment, which accentuates the requirement for sophisticated, affordable, and accessible assistive technology [3]. Studies revealed that the integration of sensors, computer vision, or audio/haptic feedback has contributed greatly to enhanced user awareness and safety [4]. Nevertheless, existing assistive technological solutions are too centered on handheld tech, which could impede natural movement as well as necessitate continuous hand usage.

Smart wearable technology, especially smart glasses, has emerged as an alternative method that is less invasive and closely resembles natural perception by humans. Smart glasses, by integrating sensors and cameras around or near the level of the user's eye, can identify and detect objects and obstacles better than canes, especially those that are around head or chest level. Vision-based intelligent systems have vast possibilities and have proven to be effective in detecting obstacles, humans, and objects in real time [5].

In practical applications, the failure to detect frontal or elevated obstacles has resulted in serious accidents, such as collisions with signboards, poles, or vehicles, especially in situations where the area is not well illuminated or is densely populated [6]. Elderly pedestrians may also fail to react quickly to avoid collisions due to impaired reflexes. Moreover, the lack of real-time feedback technology makes it difficult for traditional assist tools to alert pedestrians of potential hazards [7].

To counter the mentioned drawbacks, this project envisions the development and implementation of Smart Glasses, which will come equipped with an AI-powered vision system and audio feedback. This system features the use of a HuskyLens AI camera to enable object recognition and detection, an Arduino microcontroller for controlling purposes, and an ESP32 module for system integration and the addition of wireless connectivity features in the future. The identified obstacles and objects are relayed to the user with the help of a sound system powered by a PAM8403 audio amplifier. Every system component is chosen carefully so that it can be reliable, inexpensive, have lower power consumption, and be easily integrated. Since this project of Smart Glasses proposes an integration of embedded systems, artificial intelligence, and audio feedback in wearable form, it is intended to increase the mobility and independence of visually impaired individuals.

Advanced assistive technology has received considerable attention in improving the mobility and safety of the visually impaired and elderly individuals. The ongoing research and developments in terms of the functionality and effectiveness of this technology are significant in overcoming the difficulties of these users. Table 1 illustrates the comparison of existing assistive systems and their differences from the proposed Smart Glasses system. In recent years, conventional assistive systems like smart canes and sticks have received considerable advancements. For example, the system designed by Dey et al. in [9] uses an ultrasonic sensor and a PIC microcontroller to sense obstacles in a shorter range of 5 to 35cm for basic navigation support. Also, Gbenga in [10] presents a low-cost smart walk stick system employing ultrasonic sensors and Arduino, which has the ability to detect objects at a wider distance and provides a great starting point for other developments.

More advanced assistive tools begin to look at Internet of Things (IoT) and Global Positioning System (GPS) integration, as seen in [11], in order to support real-time location sharing, obstacle identification, as well as visual notification by means of LED displays. Other examples, as shown in [12] and [13], have incorporated IR camera capabilities, Time-of-Flight sensors, water identification, as well as facets related to user health assistance. Moreover, mobile App integration with advanced algorithms in [14] support audio notification capabilities as well as improved obstacle data, with added safety features in [15] and [16] using GSM-based notification systems, user health assistance, as well as vibrotactile feedback mechanisms.

Even with all these advances being implemented, most existing systems tend to depend on handheld canes, smartphone dependency, as well as internet access. However, with the Smart Glasses system design, the wearable technology and hands-free design are embraced by using Artificial Intelligence-vision capabilities through the HuskyLens camera and Arduino/ESP32 technology and processing capabilities that depend on giving real-time voice alerts through a speaker for enhanced awareness of both frontal and elevated obstacles without needing smartphones and internet access. Through all these considerations and implementations, it is therefore evident that there has been significant development from simple assistance technology for mobility to intelligent wearable assistance technology for the visually impaired.

Table 1 Table of comparison

Ref / Year	Title	Advantages	Disadvantages	Compared to Proposed Smart Glasses
Smart Glasses	Smart Glasses: An Intelligent Vision-Based Assistive Device for the Visually Impaired	AI-based object detection using HuskyLens; hands-free wearable design; real-time audio feedback via speaker; modular MCU–ESP32 architecture	Limited to camera field of view; no physical ground-contact feedback	Provides higher-level environmental awareness, especially for frontal and elevated obstacles
[9] 2018	Ultrasonic Sensor Based Smart Blind Stick	Low cost; simple obstacle detection using buzzer	Short detection range (5–35 cm); no object recognition; no audio guidance	Proposed system offers vision-based recognition and meaningful audio alerts
[10] 2017	Smart Walking Stick Using Ultrasonic Sensors and Arduino	Energy efficient; simple and affordable	No emergency, lighting, or intelligent processing	Smart Glasses provide intelligent, context-aware feedback rather than basic distance alerts
[11] 2022	SCBIoT: Smart Cane for Blinds Using IoT	GPS tracking; obstacle detection; LED indicators	High battery consumption; Wi-Fi dependency	Proposed system is more lightweight, wearable, and does not rely on continuous internet
[12] 2019	Smart Electronic Stick for Visually Impaired	Uses IR camera and emergency system	App-dependent; cloud/API reliance	Smart Glasses perform onboard AI processing, improving reliability and response time
[13] 2024	Smart Blind Stick	Health monitoring; water detection; GPS alert	Complex system; higher cost	Proposed system prioritizes simplicity, wearability, and real-time vision feedback
[14] 2018	Smart Blind Stick Using Arduino	Voice feedback; wearable sensors	App-reliant; no vision-based detection	Smart Glasses enhance accuracy through camera-based AI object recognition
[15] 2021	GPS and GSM Enabled Smart Blind Stick	Location tracking; vibration alerts	GSM required; complex setup	Proposed system focuses on local intelligence and audio guidance without network dependency
[16] 2018	Smart Blind Stick for Visually Impaired People	Real-time tactile feedback	Lacks audio, vision, and emergency features	Smart Glasses provide richer multimodal feedback through AI vision and audio

2. Methodology

2.1 Project Development

The Smart Glasses: An Intelligent Vision-Based Assistive Device for the Visually Impaired is developed to provide real-time obstacle detection and audio guidance to assist visually impaired users. The system integrates vision sensing, microcontroller processing, and audio feedback to ensure safe and effective navigation.

The Smart Glasses consist of several main components, including the HuskyLens camera, ESP32 microcontroller, audio amplifier, and speaker. The HuskyLens continuously scans the front area to detect obstacles and sends the detection data to the ESP32 for processing. Based on the received information, the ESP32 determines the appropriate response and generates an audio alert.

The audio alert system uses a PAM8403 amplifier to enhance the sound signal before transmitting it to the speaker, ensuring clear warning notifications. The system operates continuously after initialization and remains active until the power is turned off.

The overall workflow of the Smart Glasses is illustrated in the flowchart, which shows the sequence of system initialization, object detection, data processing, audio feedback generation, and system shutdown. This structured design ensures reliable performance and effective assistance for visually impaired users.

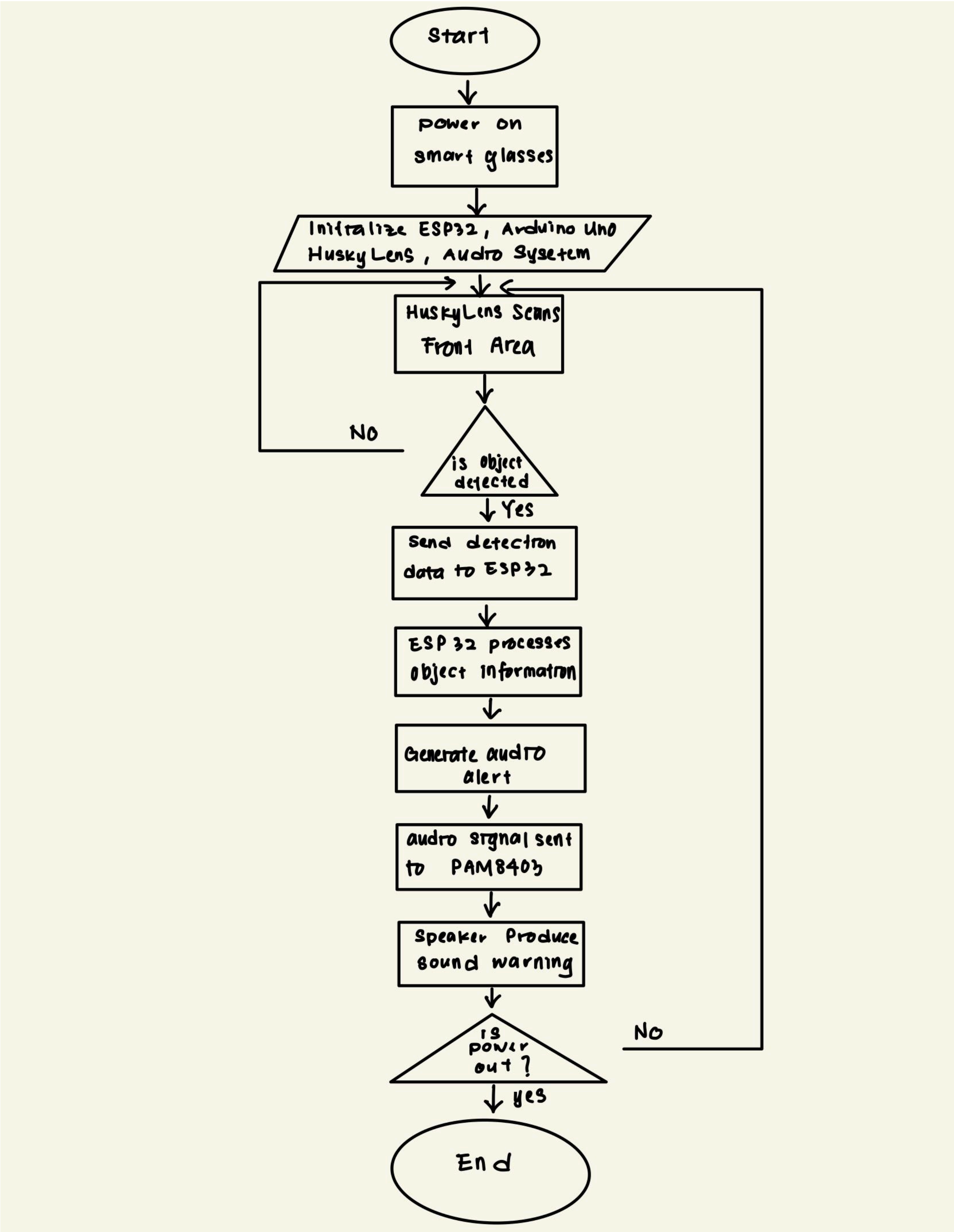

Fig 3*: Flow chart that explains all the working principles*

2.2 Schematic Diagram and Prototype Setup

Fig. 4 shows the schematic diagram of the Smart Glasses system, which integrates the HuskyLens AI camera, Arduino microcontroller, ESP32 module, PAM8403 audio amplifier, and speaker. The HuskyLens is interfaced with the Arduino for object recognition, while the ESP32 is connected for system integration and future wireless functionality. Fig. 5 illustrates the physical prototype setup, where all components are mounted onto a wearable glasses frame with the camera positioned at the front and the audio output placed near the ear, forming a compact and hands-free assistive device.

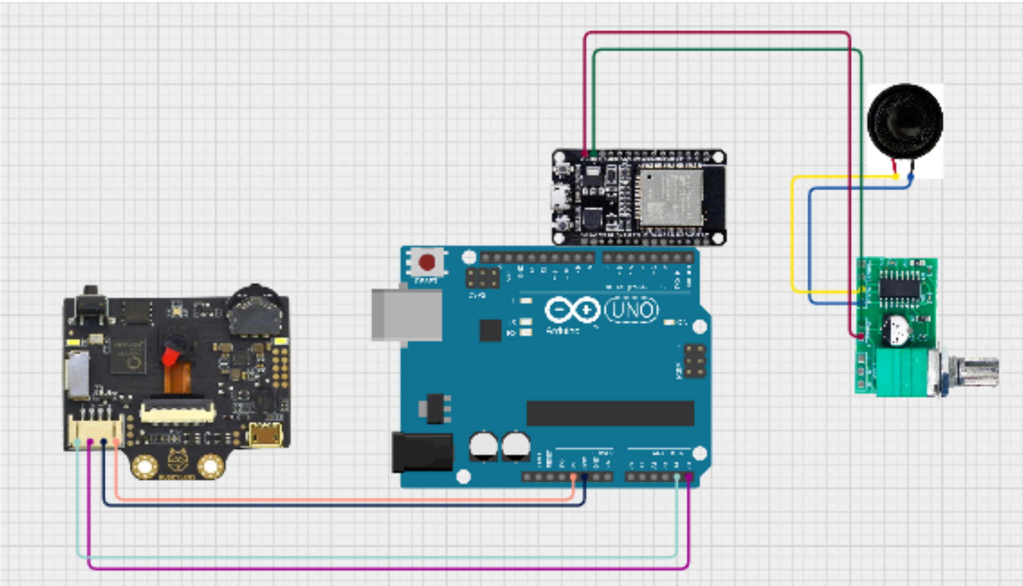

Fig. 4 Schematic diagram of the Smart Glasses system

Fig. 5b Smart Glasses prototype

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